Exact Small Parsimony for Natural Genomes

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August 31, 2026

Marker, Chromosome, Genome

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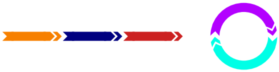

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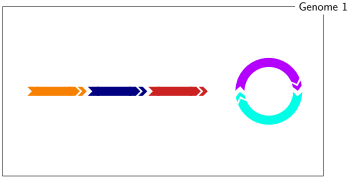

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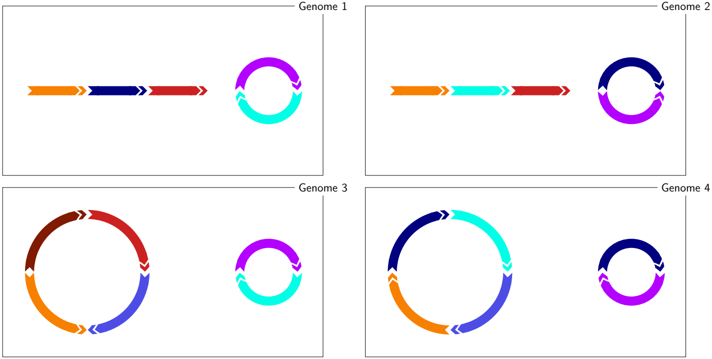

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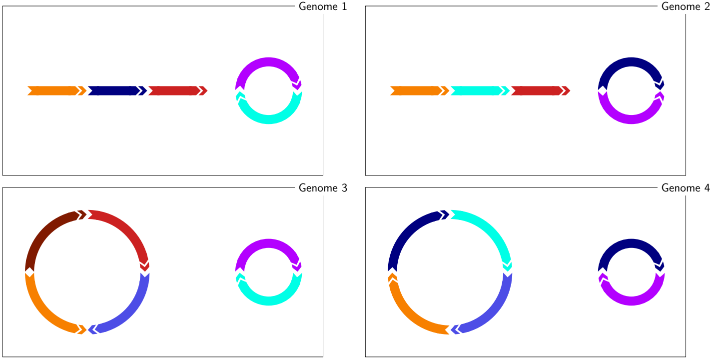
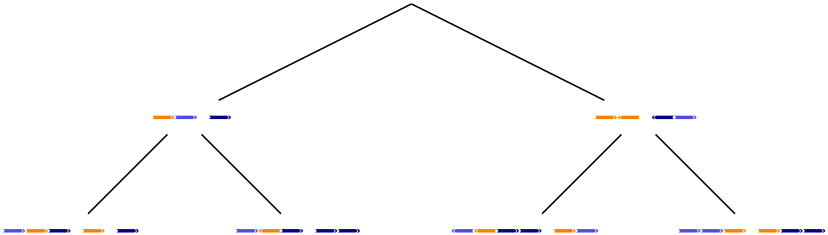


Natural genomes model: <u>Unrestricted</u> copy number.

Small Parsimony for Natural Genomes



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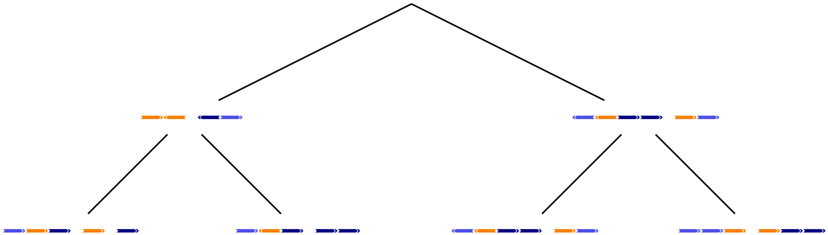

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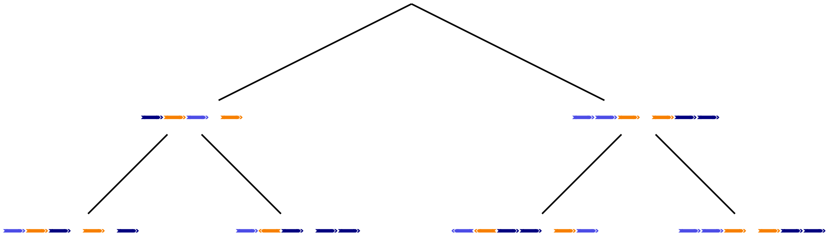

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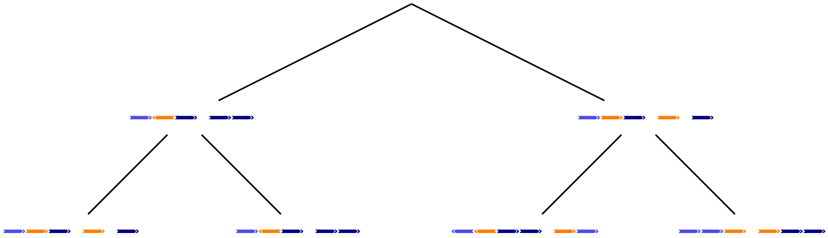

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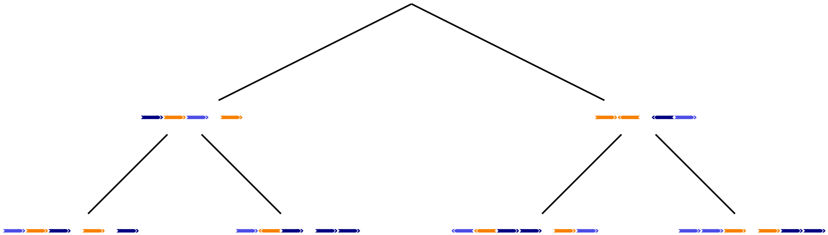

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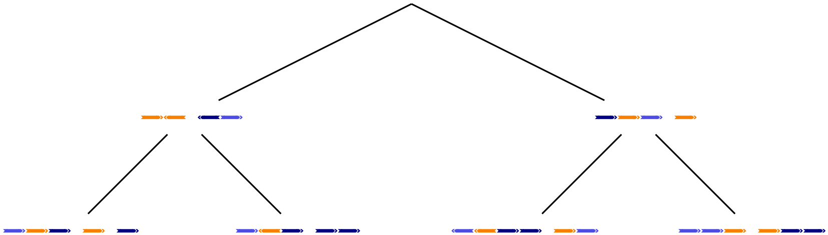

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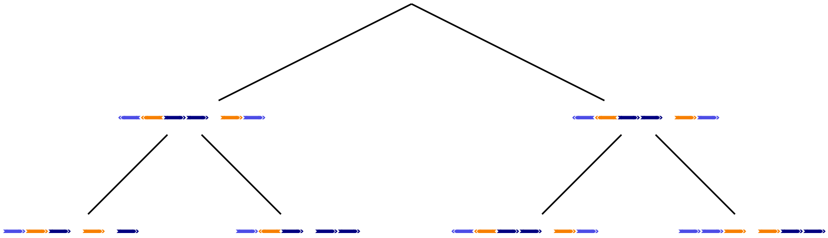
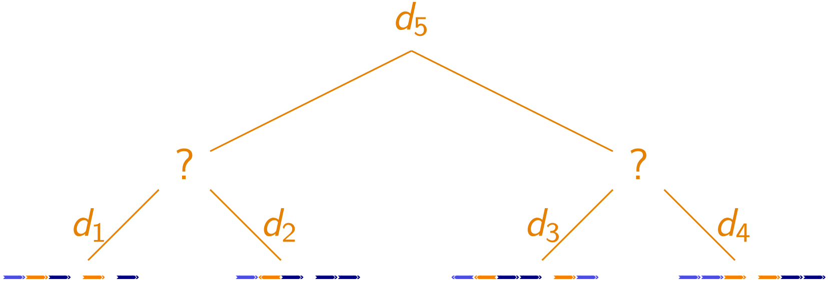
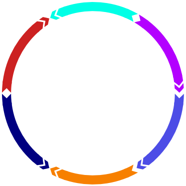


Small Parsimony for Natural Genomes

Minimize $d_1 + d_2 + d_3 + d_4 + d_5$



Model: Double-Cut-And-Join



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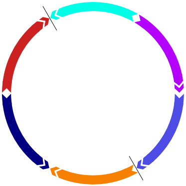

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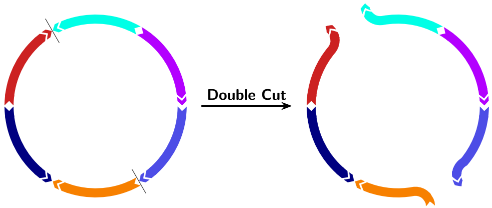

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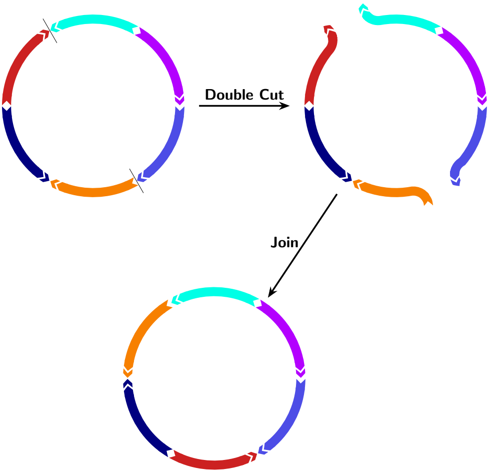

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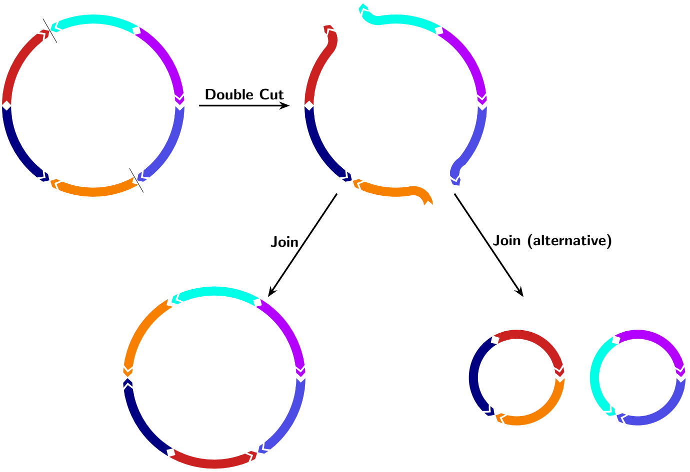
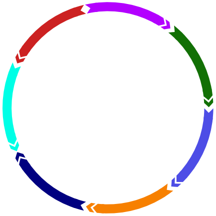


Insertions and Deletions



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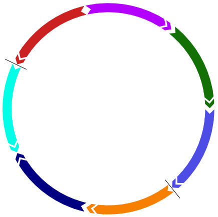

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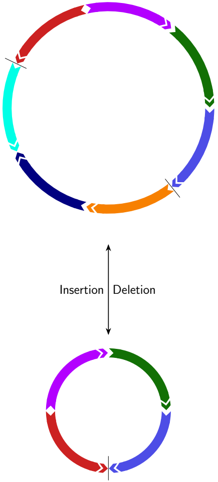
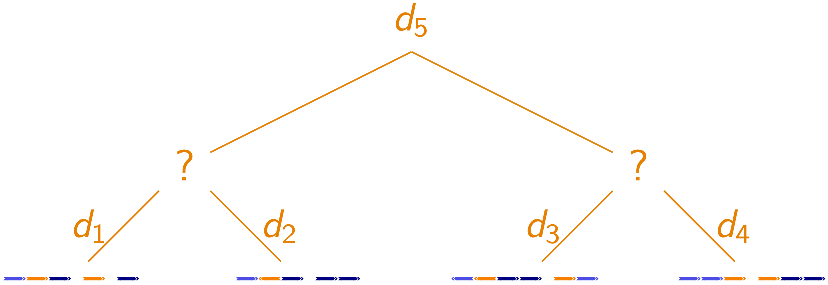


Small Parsimony for Natural Genomes

Minimize $d_1 + d_2 + d_3 + d_4 + d_5$



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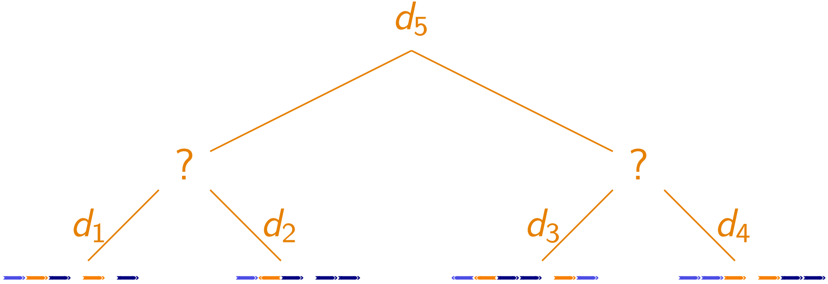
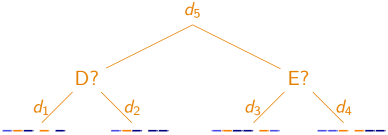


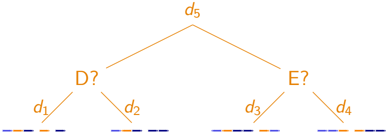
d_1, d_2, d_3, d_4, d_5: (Minimum number of) DCJ and indel operations between the genomes.

Small Parsimony for Natural Genomes via Degenerate Genomes (SPP-DCJ)



NP Hard

Small Parsimony for Natural Genomes via Degenerate Genomes (SPP-DCJ)



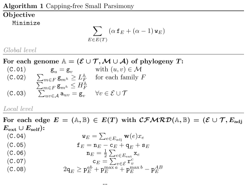
NP Hard

Heuristic

Degenerate Genomes

+ *A*, *B*, *C*

Integer Linear Program (ILP)



Algorithm 1 Capping-free Small Parsimony

Objective

Minimize

$$\sum_{E \in E(T)} (\alpha\, \mathtt{f}_E + (\alpha - 1)\, \mathtt{w}_E)$$

Global level

For each genome $\mathbb{A} = (\mathcal{E} \cup \mathcal{T}, \mathcal{M} \cup \mathcal{A})$ of phylogeny T:

(C.01)	$\mathtt{g}_u = \mathtt{g}_v$	with $(u, v) \in \mathcal{M}$
(C.02)	$\sum_{m \in F} \mathtt{g}_{m^h} \geq L_F^{\mathbb{A}}$	for each family F
	$\sum_{m \in F} \mathtt{g}_{m^h} \leq H_F^{\mathbb{A}}$	
(C.03)	$\sum_{uv \in \mathcal{A}} \mathtt{a}_{uv} = \mathtt{g}_v$	$\forall v \in \mathcal{E} \cup \mathcal{T}$

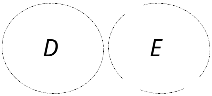
Local level

For each edge $E = (\mathbb{A}, \mathbb{B}) \in E(T)$ with $\mathcal{CFMRD}(\mathbb{A}, \mathbb{B}) = (\mathcal{E} \cup \mathcal{T}, E_{\text{adj}}$ $E_{\text{ext}} \cup E_{\text{self}})$:

(C.04)	$\mathtt{w}_E = \sum_{e \in E_{\text{adj}}} \mathbf{w}(e) x_e$
(C.05)	$\mathtt{f}_E = \mathtt{n}_E - \mathtt{c}_E + \mathtt{q}_E + \mathtt{s}_E$
(C.06)	$\mathtt{n}_E = \frac{1}{2} \sum_{e \in E_{\text{ext}}} \mathtt{x}_e$
(C.07)	$\mathtt{c}_E = \sum_{v \in \mathcal{E}} \mathtt{r}_v^c$
(C.08)	$2\mathtt{q}_E \geq \mathtt{p}_E^{ab} + \mathtt{p}_E^{\max a} + \mathtt{p}_E^{\max b} - \mathtt{p}_E^{AB}$

...

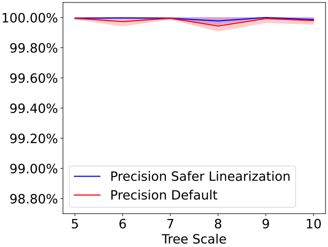
Derived Genomes

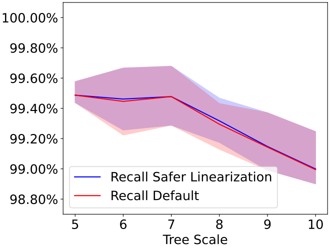


Adapted from (Doerr, Chauve, 2021).

SPP-DCJ – the Good

On simulated data (ZOMBI+pre-filtering with DeCoSTAR), *if the ground truth adjacencies are represented in the degenerate genome*:



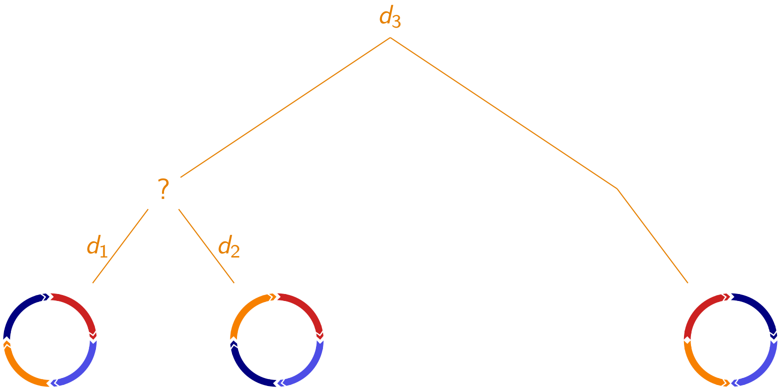


(WABI 2024)

SPP-DCJ – the Bad

- Good pre-filtering (DeCoSTAR) requires gene trees 
- Is not exact 
- Biases the results by confidence in adjacencies
- Requires given copy numbers

SPP-DCJ – the Ugly



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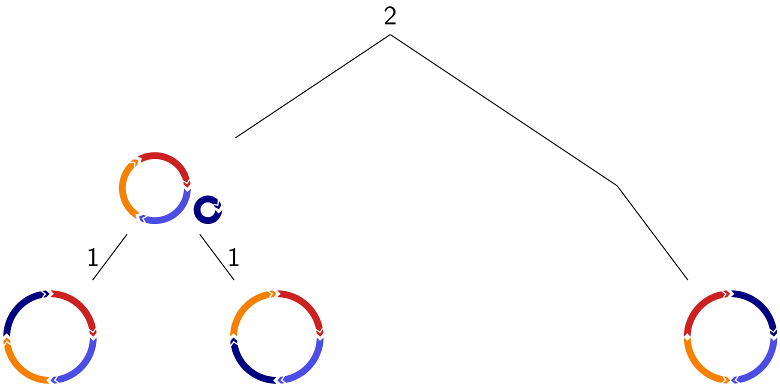

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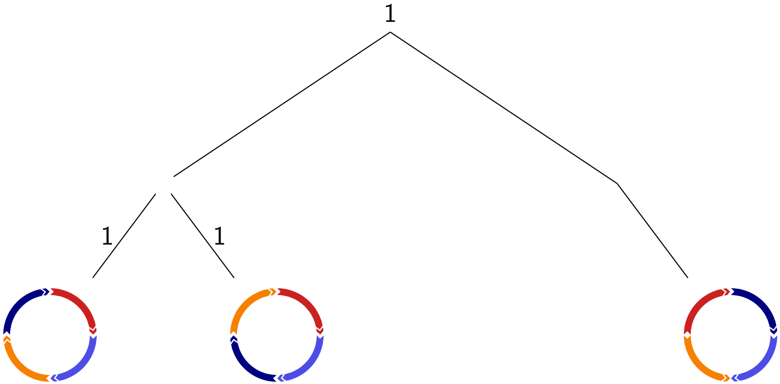
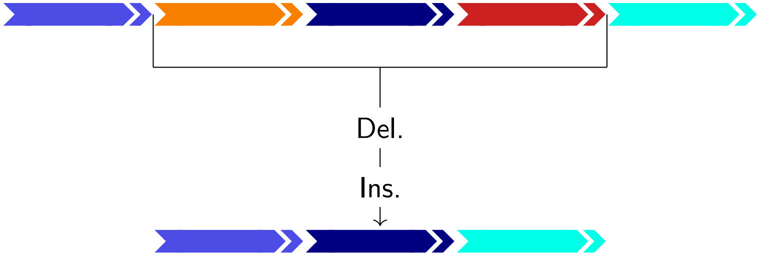


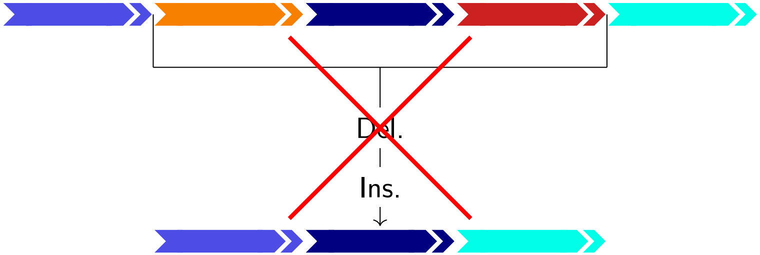
DCJ-indel distance



Original model:

- Delete only overrepresented markers
- Insert only underrepresented markers
- Cost 1 for any indel

DCJ-indel distance



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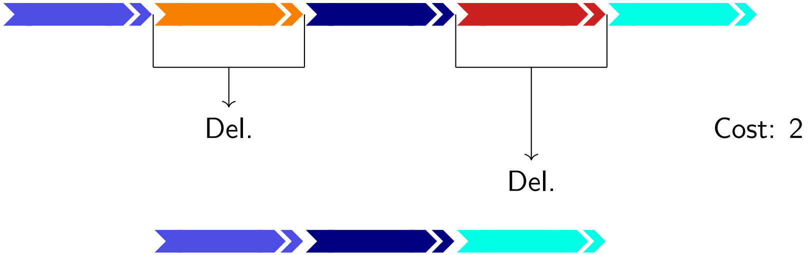

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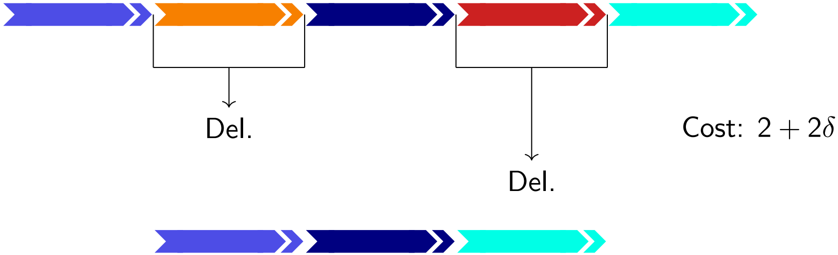
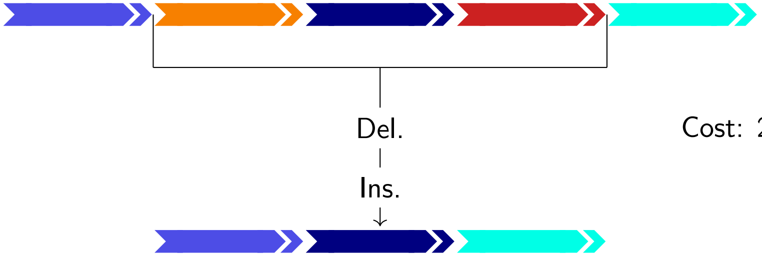


Our model:

- Any indel is allowed
- Cost $1 + \delta\ell$ for an indel of length ℓ

DCJ-indel distance



Cost: $2 + 4\delta$

Our model:

- Any indel is allowed
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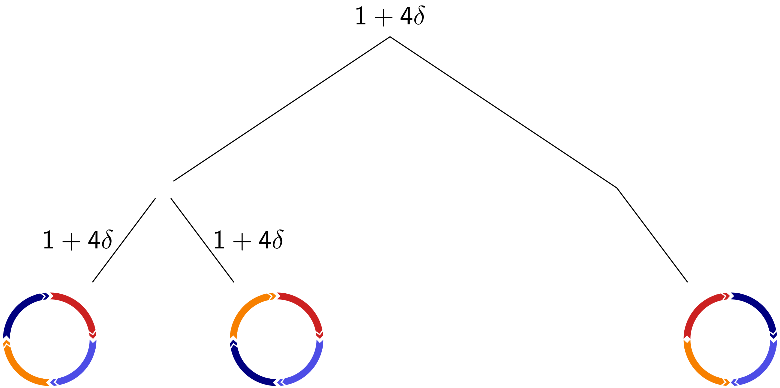
SPP-DCJ – A Solution



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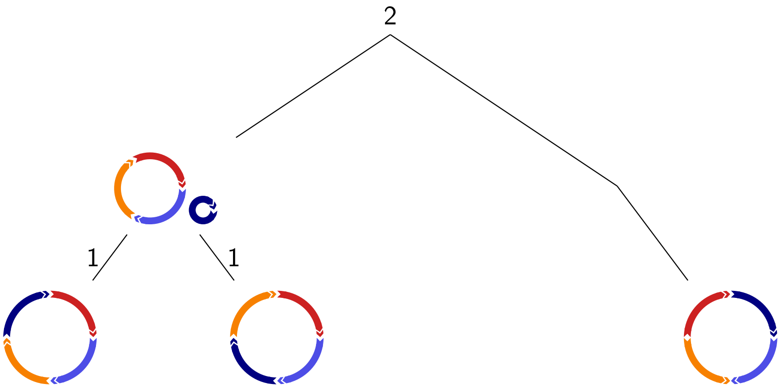

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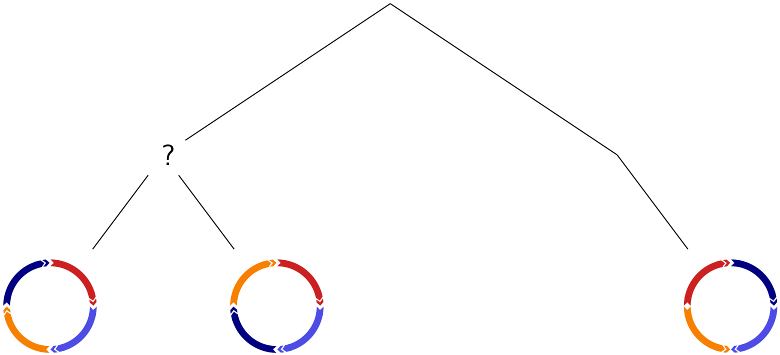
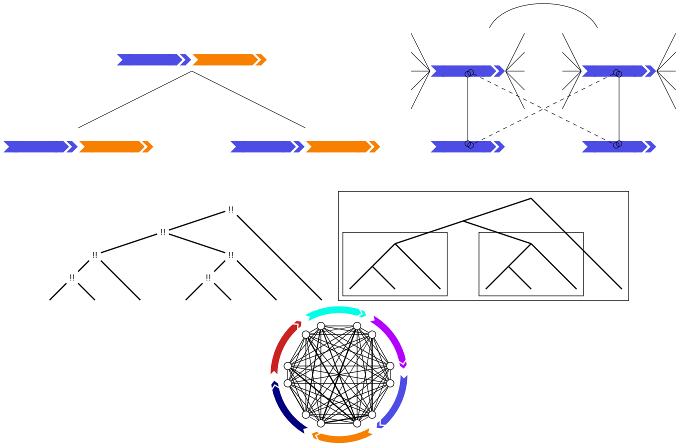


- Delete/Insert only overrepresented/underrepresented: $\delta \to \infty$

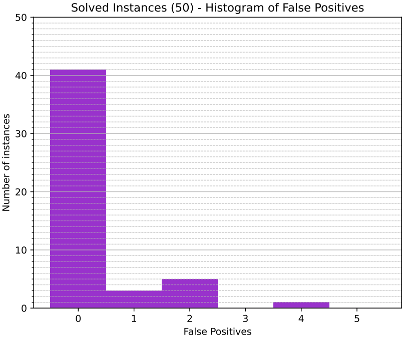
$\rightarrow$ dominant δ-term allows to infer copy number ranges before solving ILP

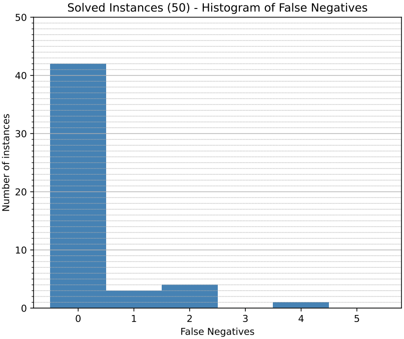
- Wagner Parsimony – polynomial time algorithm (Swofford-Madison 1987)

Theory and Solver Tricks

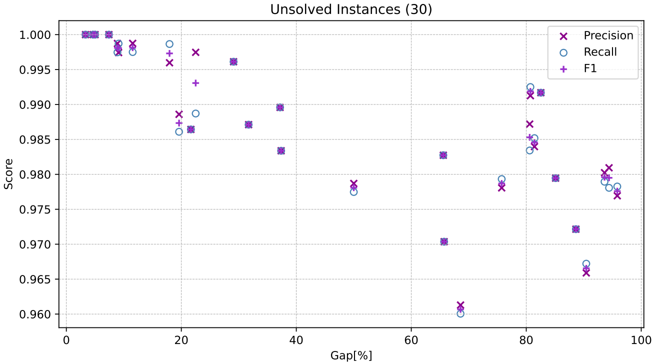


Reconstruction Quality - Exact

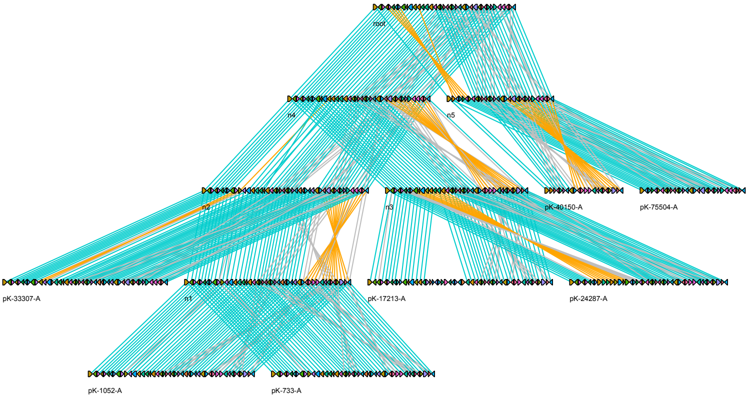




Reconstruction Quality - Approximate



Reconstruction Quality - Plasmids



SPP - Conclusion and Long Term Perspectives

Conclusion:

- SPP-exact needs only extant genomes
- Fixed "free lunch" DCJ-indel via δ-term
- Reducing the solution space $\rightarrow$ large impact on SPP
- Biologically meaningful instances (plasmids) solvable close to optimal

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Future work:

- Support larger problem instances
  - Reduce to sub-problems (common intervals?)
- $\delta < \infty$?
- Connection with Duplication-Transfer-Loss Small (DTL) Parsimony (horizontal transfers, chromosome duplication,...)
- Large Parsimony problem (no given tree)

Thank you!